

《常微分方程》

2018-2019 学年第二学期期末考试 A 卷

一、求解下列微分方程。(每题 10 分)

1. $xy' - y = \sqrt{x^2 - y^2}$

2. $(x^2y^2 + 2x)dy - 2ydx = 0$ 满足初始条件 $y(1) = 0$ 的特解

3. $yy'' - (y')^2 = y^2y'$

$$4. (x-1)y'' - xy' + y = -(x-1)^2 e^x$$

$$5. x^2 y''' - xy'' - 3y' = \frac{4}{x}, x > 0$$

二、已知 $f(x)$ 二阶连续可导，且 $f(x) = e^{-x} + \int_0^x tf(x-t)dt$ ，求 $f(x)$ 。(10 分)

三、已知 $y_1(x) = \cos x + x$ 和 $y_2(x) = 2 \sin x + x$ 是方程 $\frac{d^2 y}{dx^2} + a \frac{dy}{dx} + by = f(x)$ 的特解，求 $a, b, f(x)$ 表达式及方程的通解。(10 分)

四、求下列微分方程。(每题 10 分)

$$1. \begin{cases} \frac{dx}{dt} = -4x - y \\ \frac{dy}{dt} = -x + 4y - z \\ \frac{dz}{dt} = -y + 4z \end{cases}$$

$$2. \begin{cases} \frac{dx}{dt} = -2y + e^t \\ \frac{dy}{dt} = 2x + 1 + t \end{cases}$$

2018-2019 学年第二学期期末考试 A 卷参考答案

一、

1、【解析】

$$xy' - y = \sqrt{x^2 - y^2}$$

左右两侧各除以 $x(x \neq 0)$, 得 $y' - \frac{y}{x} = \sqrt{1 - (\frac{y}{x})^2}$,

$$\text{令 } u = \frac{y}{x}, \text{ 则 } y = ux, \frac{dy}{dx} = x \frac{du}{dx} + u, \text{ 代入得 } x \frac{du}{dx} + u - u = \sqrt{1 - u^2},$$

$$x \frac{du}{dx} = \sqrt{1 - u^2}, \text{ 分离变量得, } \frac{dx}{x} = \frac{du}{\sqrt{1 - u^2}},$$

$$\text{分别积分 } \int \frac{dx}{x} = \int \frac{du}{\sqrt{1 - u^2}} \text{ 得到, } \arcsin u = \ln |x| + C,$$

即是 $\arcsin \frac{y}{x} = \ln |x| + C, y = x \sin(\ln |x| + C), C$ 是任意常数。

【考点延伸】分离变量法的应用

2、【解析】 $(x^2 y^2 + 2x)dy - 2ydx = 0$

$$\text{令 } u = \frac{x}{y}, \text{ 则 } x = uy, \frac{dx}{dy} = y \frac{du}{dy} + u, \text{ 原式化为 } \frac{dx}{dy} = \frac{x^2 y}{2} + \frac{x}{y},$$

$$\text{即 } y \frac{du}{dx} + u = \frac{u^2 y^3}{2} + u \text{ 分离变量得, } y^2 dy = 2 \frac{du}{u^2},$$

$$\text{分别积分 } \int y^2 dy = \int 2 \frac{du}{u^2}, \text{ 得到, } -\frac{2}{u} = \frac{y^3}{3} + C,$$

$$\text{则 } -\frac{2y}{x} = \frac{y^3}{3} + C, \text{ 即 } 0 = \frac{y^3}{3} + \frac{2y}{x} + C, C \text{ 是任意常数,}$$

将 $y(1) = 0$ 代入求得特解 $C = 0$,

$$\text{则特解为 } 0 = \frac{y^3}{3} + \frac{2y}{x} (y \neq 0).$$

【考点延伸】分离变量法的应用

3、【解析】 $yy'' - (y')^2 = y^2 y'$

$$\text{令 } y' = p, \text{ 则 } y'' = \frac{dp}{dy} \frac{dy}{dx} = p \frac{dp}{dx},$$

$$\text{代入原式得 } yp \frac{dp}{dy} - p^2 = y^2 p \Rightarrow p \frac{dp}{dy} = \frac{y^2 p + p^2}{y} = yp + \frac{p^2}{y},$$

$$\text{若 } p \neq 0, \frac{dp}{dy} = y + \frac{p}{y}, \text{ 方程为 } \frac{dp}{dy} - \frac{p}{y} = y$$

$$\text{通解为 } p = e^{-\int \frac{1}{y} dy} \left(\int y e^{\int \frac{1}{y} dy} dy + C \right),$$

$$\text{得 } p = y^2 + cy = \frac{dy}{dx}, \text{ 两端积分得 } \int \frac{dy}{y^2 + cy} = \int 1 dx$$

$$-\frac{1}{C_1} \ln \left| \frac{y+C_1}{-y} \right| + C_2 = x, C_1, C_2 \text{ 为任意常数.}$$

【考点延伸】高阶微分的处理方法和通解公式

4、【正解】 $(x-1)y'' - xy' + y = -(x-1)^2 e^x$

二阶变系数线性常微分的齐次方程为

$$(x-1)y'' - xy' + y = 0,$$

$$f(x) = x-1, p(x) = -x, q(x) = 1, f(x) + p(x) + q(x) = 0,$$

则有一特解为 $y_1 = e^x$,

$$\text{使用刘维尔公式得到 } y_2 = y_1 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx,$$

$$P(x) = \frac{p(x)}{f(x)} = -\frac{x}{x-1}, \text{ 得到 } y_2 = -x, \text{ 则齐次方程的通解为}$$

$$Y = C_1 e^x + C_2 (-x), \text{ 令非齐次的通解为}$$

$$y = V_1(x)e^x + V_2(x)(-x), \text{ 即是}$$

$$\begin{cases} e^x V_1' + (-x)V_2' = 0 \\ e^x V_1' - V_2' = \frac{g(x)}{f(x)} = -(x-1)e^x \end{cases}, \text{ 解得 } \begin{cases} V_1' = -x \\ V_2' = -e^x \end{cases}$$

$$\text{积分得 } \begin{cases} V_1 = -\frac{x^2}{2} + C_1 \\ V_2 = -e^x + C_2 \end{cases}$$

则通解为

$$y = \left(-\frac{x^2}{2} + C_1\right)e^x + (-e^x + C_2)(-x) = -\frac{x^2}{2}e^x + C_1e^x + xe^x - C_2x.$$

【考点延伸】刘维尔公式的应用

5、【解析】 $x^2 y''' - xy'' - 3y' = \frac{4}{x}, x > 0$

令 $x = e^t$, 则 $t = \ln x, \frac{dt}{dx} = \frac{1}{x}$,

原式化为, $x^3 y''' - x^2 y'' - 3xy' = 4$,

引入微分算子 $D = \frac{d}{dt}, (D^3 - 3D^2 + 2D)y - (D^2 - D)y - 3Dy = 4$,

$(D^3 - 4D^2)y = 4$, 即是 $y''' - 4y'' = 4$.

$\lambda^3 - 4\lambda^2 = 0, \lambda_1 = 4, \lambda_2 = \lambda_3 = 0, y = C_1 e^{4x} + e^{0x}(C_2 x + C_3)$,

特解设为 $y_0 = Ax^2, y_0' = 2Ax, y_0'' = 2A, y_0''' = 0$.

则 $0 - 4 \times 2A = 4, A = -\frac{1}{2}$, 得到 $y_0 = -\frac{1}{2}x^2$.

所以解为 $y = C_1 e^{4x} - \frac{1}{2}x^2 + C_2 x + C_3$.

【考点延伸】解的组成: 特解和齐次解的求法

二、

【解析】

$f(x) = e^{-x} + \int_0^x t f(x-t) dt$,

求导, $f'(x) = -e^{-x} + [\int_0^x t f(x-t) dt]'$.

令 $x-t=u$, 得 $\int_0^u (x-u)f(u)du = x \int_0^u f(u)du - \int_0^u f(u)du$

则求导 $xf(x) + \int_0^u f(u)du - xf(x) = \int_0^u f(u)du$.

$f'(x) = -e^{-x} + \int_0^u f(u)du, f''(x) = e^{-x} + f(x)$,

即是 $y'' - y = e^{-x}$

齐次方程的特征方程为 $\lambda^2 - 1 = 0, \lambda = \pm 1$, 则通解为 $y = C_1 e^x + C_2 e^{-x}$,

特解为 $y^* = A e^{-x} x, y'' = A e^{-x}(x-2), A e^{-x}(x-2) - A x e^{-x} = e^{-x}$,

$A = -\frac{1}{2}$, 则 $y^* = -\frac{1}{2} x e^{-x}$.

方程的解为 $y = f(x) = C_1 e^x + C_2 e^{-x} - \frac{1}{2} x e^{-x}$.

【考点延伸】塑造微分方程和特征方程求解

三、

【解析】

将 y_1, y_2 分别代入方程, 得

$$\begin{cases} -2\cos x + a(1 - \sin x) + b(\cos x + x) = f(x) \\ -2\sin x + a(2\cos x + 1) + b(2\sin x) = f'(x) \end{cases}$$

解得 $a=0, b=1, f(x)=x$,

则通解为 $y = C_1 y_1 + C_2 y_2 + y^*$.

该齐次方程为 $y'' + y = 0, \lambda^2 + 1 = 0, \lambda = \pm i$,

则通解为 $y = e^{0x} (C_1 \cos x + C_2 \sin x)$,

又特解已知, 不妨令 $y^* = y_1$,

非齐次方程的通解为 $y = C_1 \cos x + C_2 \sin x + \cos x + x$.

【考点延伸】解的含义和解的构成

四、【解析】

1、

$$\begin{cases} \frac{dx}{dt} = -4x - y & (1) \\ \frac{dy}{dt} = -x + 4y - z & (2) \\ \frac{dz}{dt} = -y + 4z & (3) \end{cases}$$

$$y'' = (-x + 4y - z)' = 4y' - (x' + z') = 4y' + 2y + 4x - 4z(4),$$

联合(2),(4), 解得 $8x = y'' - 8y' + 14y(5)$, 即是 $2x' = -y'' + 8y' - 16y$.

又(5)求导得, $8x' = y''' - 8y'' + 14y' = 4(-y'' + 8y' - 16y)$,

$$\text{即 } y''' - 4y'' - 18y' + 64y = 0.$$

特征方程为 $\lambda^3 - 4\lambda^2 - 18\lambda + 64 = 0$, 设其解为 $\lambda_1, \lambda_2, \lambda_3$,

所以, 齐次解为 $y = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} + C_3 e^{\lambda_3 t}$,

则 $x' + 4x = -y$, 那么 $x = C_4 e^{-4t} + D_1 e^{\lambda_1 t} + D_2 e^{\lambda_2 t} + D_3 e^{\lambda_3 t}$, 特解代入得到

$$D_1 = \frac{C_1}{4 + \lambda_1}, D_2 = \frac{C_2}{4 + \lambda_2}, D_3 = \frac{C_3}{4 + \lambda_3}.$$

同理可得 $z' - 4z = -y$, 那么 $z = C_5 e^{4t} + \frac{C_1}{\lambda_1 - 4} e^{\lambda_1 t} + \frac{C_2}{\lambda_2 - 4} e^{\lambda_2 t} + \frac{C_3}{\lambda_3 - 4} e^{\lambda_3 t}$.

所以

$$y = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t} + C_3 e^{\lambda_3 t},$$

$$x = C_4 e^{-4t} + \frac{C_1}{4 + \lambda_1} e^{\lambda_1 t} + \frac{C_2}{4 + \lambda_2} e^{\lambda_2 t} + \frac{C_3}{4 + \lambda_3} e^{\lambda_3 t},$$

$$z = C_5 e^{4t} + \frac{C_1}{\lambda_1 - 4} e^{\lambda_1 t} + \frac{C_2}{\lambda_2 - 4} e^{\lambda_2 t} + \frac{C_3}{\lambda_3 - 4} e^{\lambda_3 t},$$

其中 $\lambda_1, \lambda_2, \lambda_3$ 为特征方程为 $\lambda^3 - 4\lambda^2 - 18\lambda + 64 = 0$ 的不同的根.

$$2、\begin{cases} \frac{dx}{dt} = -2y + e' & (1) \\ \frac{dy}{dt} = 2x + 1 + t & (2) \end{cases}$$

由(1)得, $y = -\frac{1}{2}\frac{dx}{dt} + \frac{1}{2}e'$, 对 t 求导, $\frac{dy}{dt} = -\frac{1}{2}\frac{d^2x}{dt^2} + \frac{1}{2}e' & (3)$,

将(3)代入(2), $-\frac{1}{2}\frac{d^2x}{dt^2} + \frac{1}{2}e' = 2x + 1 + t$, 即是 $x'' + 4x = e' - 2 - 2t$.

特征方程 $\lambda^2 + 4 = 0$, $\lambda = \pm 2i$, 通解 $x = C_1 \cos 2t + C_2 \sin 2t + x^*$.

设 $x^* = Ae^t + B + Ct$, $(x^*)'' = Ae^t$,

代入 $Ae^t + 4Ae^t + 4B + 4Ct = e' - 2 - 2t$, 得

$$A = \frac{1}{5}, B = -\frac{1}{2}, C = -\frac{1}{2}.$$

$$\text{则 } x = C_1 \cos 2t + C_2 \sin 2t + \frac{1}{5}e^t - \frac{1}{2}t - \frac{1}{2}.$$

$$\frac{dx}{dt} = -2C_1 \sin 2t + 2C_2 \cos 2t + \frac{1}{5}e^t - \frac{1}{2},$$

$$y = -\frac{1}{2}\frac{dx}{dt} + \frac{1}{2}e' = -\frac{1}{2}(-2C_1 \sin 2t + 2C_2 \cos 2t + \frac{1}{5}e^t - \frac{1}{2}) + \frac{1}{2}e'$$

$$= C_1 \sin 2t - C_2 \cos 2t + \frac{2}{5}e' + \frac{1}{4}.$$

【考点延伸】常微分方程组的求解

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